

# BADG 2: A Perfect Game

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## Abstract

The earlier (b. a. d. g.) paper focused on a near-optimal strategy for maximizing expected value in gameplay. This paper focuses on a different question: what is the probability of playing a perfect game? Using standard recursive calculation, we calculate the probabilities with the out-of-box game setup. Then we venture into a related: What's the expected win probability with  $n$  standard dice of side count  $s$  as  $n \rightarrow \infty$ ? We give poor bounds and prove convergence.

**Theorem 0.1.** *The expected chance of a perfect game with  $r$  standard dice of  $s$  sides each, given  $p := \frac{1}{s}$ , is  $P(r) = \sum_{k=1}^r \binom{r}{k} p^k q^{r-k} P(r-k)$ , with  $P(0)$  defined as 1.*

*Proof:* This is a clear application of the binomial theorem.

Bounding this quantity will be the principal thrust of this paper.

**Lemma 0.2** (Easy Cases Lemma). *If  $p = 0$ ,  $P(r) = 0$ . If  $p = 1$ ,  $P(r) = 1$ . If  $s = 2$ , for any  $r > 0$ ,  $P(r) = \frac{1}{2}$ .*

*Proof:* Though we won't be flipping infinite ( $p = 0$ ) or 1-sided ( $p = 1$ ), dice, these algebraic identities are obvious (can't succeed, can't fail, respectively) and will be useful later.

**Base case for  $s=2$ :**  $P(1) = \frac{1}{2}$  since a perfect game occurs if and only if the coin flips 2 instead of 1.

**Inductive case for  $s=2$ :** Assuming  $P(k) = \frac{1}{2}$  for all  $0 < k < r$ , and  $P(0) = 1$ :

$$\sum_{k=1}^{r-1} \binom{r}{k} p^k q^{r-k} P(r-k) + \binom{r}{r} p^r q^0 P(0) = \quad (1)$$

$$(2^r - 2) \frac{1}{2^r} \left(\frac{1}{2}\right) + \frac{1}{2^r} 2 \left(\frac{1}{2}\right) = \quad (2)$$

$$(2^r) \frac{1}{2^r} \left(\frac{1}{2}\right) = \frac{1}{2} \quad (3)$$

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- Extremely Lazy Lower bound (closed form)
- Wrong attempt:  $F(t) < F(t - 1)$ ?
- $F(n)$  exhibits provably asymptotic behavior

**Theorem 0.3.** *For an unbounded set of  $r > 0$  standard  $s$ -sided dice,  $s \geq 2$ , the probability  $P(r)$  of getting a perfect game has a lower bound of  $\exp(-\frac{q}{(1-q)^2})$ , if we define  $q := 1 - \frac{1}{s}$ .*

*Proof:*

- First, while at stage  $k$ , where  $k$  dice remain, the probability  $P_s(k)$  of continuing to some stage  $j < k$  is clearly  $1 - q^k$ .
- The probability  $P(k)$  of passing all such stages is greater than or equal to  $P^* = \prod_{i=1}^k P_s(k)$ , since a successful traversal may involve skipping one or more stages, and  $P^*$  assumes we no stages.
- Therefore,  $P(k) \geq \prod_{i=1}^k P_s(k) = \prod_{i=1}^k (1 - q^k)$ .
- This means  $\log(P(k)) \geq \sum_{i=1}^k \log(P_s(k)) = \sum_{i=1}^k \log(1 - q^k)$ .

This  $P(k)$  is apparently defined as the **q-Pochhammer symbol**  $(q; q)_\infty = \prod_{i=1}^k (1 - q^k)$ , with known convergence properties. I'll instead attempt to tackle convergence without the aid of these theorems.

Often,  $\log(1 - z)$  is approximated by  $-z$  near zero, but we are most concerned with  $z$  near one, since  $1 - z$  approaches 0, and  $\log(1 - z)$  explodes. Therefore, let's look at the expansion of  $\log(1 - z)$ :

- Identity 1:  $f(z) = -\frac{1}{1-z} = -1 - z - z^2 - \dots$
- Identity 2:  $\int f(z) = \log(1 - z) = -z - \frac{z^2}{2} - \frac{z^3}{3} - \dots$

So when summing  $\sum_{i=1}^k \log(1 - q^i)$ , we can instead sum

$$\log(P(k)) = \log(1 - q^1) + \log(1 - q^2) + \log(1 - q^3) + \dots \quad (4)$$

$$\log(P(k)) = (-q - \frac{q^2}{2} - \frac{q^3}{3} + \dots) + (-q^2 - \frac{q^4}{2} - \frac{q^6}{3} + \dots) + (-q^3 - \frac{q^6}{2} - \frac{q^9}{3} + \dots) + \dots \quad (5)$$

$$> (-q - q^2 - q^3 + \dots) + (-q^2 - q^4 - q^6 + \dots) + (-q^3 - q^6 - q^9 + \dots) + \dots \quad (6)$$

$$= \frac{-q}{1-q} + \frac{-q^2}{1-q^2} + \frac{-q^3}{1-q^3} + \dots \quad (7)$$

$$> \frac{-q}{1-q} + \frac{-q^2}{1-q} + \frac{-q^3}{1-q} + \dots \quad (8)$$

$$= \frac{1}{1-q} [-q - q^2 - q^3 - \dots] \quad (9)$$

$$= \frac{-q}{(1-q)^2} \quad (10)$$

- (2) follows from (1) by substitution using Identity1 above.
- (3) follows from (2) since the denominators of (2) are larger.
- (4) follows from (3) when dividing each sequence by  $-q^k$  and applying identity 2.
- (5) follows from (4) since the denominators of (4) are larger.
- (6) and (7) follow from (5) by dividing out  $\frac{1}{1-q}$  and applying Identity 1.

Since  $\log(P(k)) > \frac{-q}{(1-q)^2}$ ,  $P(k) > \exp(\frac{-q}{(1-q)^2})$ .

| s  | p              | q              | bound                                    |
|----|----------------|----------------|------------------------------------------|
| 2  | $\frac{1}{2}$  | $\frac{1}{2}$  | $\exp(-2) \approx .135$                  |
| 3  | $\frac{1}{3}$  | $\frac{1}{3}$  | $\exp(-6) \approx .00248$                |
| 4  | $\frac{1}{4}$  | $\frac{1}{4}$  | $\exp(-12) \approx 6.14 \times 10^{-6}$  |
| 10 | $\frac{1}{10}$ | $\frac{1}{10}$ | $\exp(-90) \approx 8.19 \times 10^{-40}$ |

Table 1: Very loose lower bounds on  $P(k = \infty)$

This is a thoroughly unsatisfying lower bound, as we know  $P(2) = \frac{1}{2}$ .

However, this does provide a positive lower bound, and brings up this interesting fact: **Though increasing the number of sides of the dice can drive  $P(k)$  to zero, increasing  $k$  to  $\infty$  will never set  $P(k)$  to zero.**

If we can couple this with the proof of the next theorem, we can state that  $P(k)$  has an asymptotic limit as  $k \rightarrow \infty$  given some  $q$ .

**Calculating by hand:**

- $P(0) = 1$
- $P(1) = p$
- $P(2) = p^2(1 + 2q) = p^2(3 - 2p)$
- $P(3) = p^3(13 - 27p + 21p^2 - 6p^3)$
- $P(4) = p^4(75 - 316p + 606p^2 - 664p^3 + 432p^4 - 156p^5 + 24p^6)$
- $P(1) - P(0) = p - 1 = -(1 - p)$
- $P(2) - P(1) = p^2(3 - 2p) - p = (-1)p(1 - 2p)(1 - p)$
- $P(3) - P(2) = (-1)3p^2(1 - p)^3(1 - 2p)$
- $P(4) - P(3) = -p^3(1 - 2p)(p - 1)^2(12p^4 - 48p^3 + 72p^2 - 50p + 13)$

**Theorem 0.4** (Incorrect Blind Alley:  $P(n) < P(n - 1)$ ). *If  $p < \frac{1}{2}$ ,  $F(n - 1) > F(n)$  for  $n > 1$ .*

- TODO: Show calculation of various functions directly.
- Show calculation of  $p = \frac{1}{2} - \epsilon$ ,  $q = \frac{1}{2} - \epsilon$  with  $p = 0.495$ .
- Show interesting observations: growth of polynomial
- $R(t)$  is starting at position  $t$ , throwing  $t - 1$  dice.
- $P(t) = p(1 + q^{t-1})P(t - 1) + qR(t) = pP(t - 1) + pq^{t-1}P(t - 1) + qR(t)$
- If  $qR(t) < qP(t - 1) - pq^{t-1}P(t - 1)$ , same as  $P(t - 1) - R(t) > pq^{t-2}P(t - 1)$ , then  $P(t) < P(t - 1)$ .
- In the special case  $q = p = \frac{1}{2}$ , then  $P(t - 1) - R(t) = p^{t-1}(1) - p^{t-1} \cdot p + \text{other zero factors} = p^t = pq^{t-2} \cdot \frac{1}{2}$ .
- So we're satisfied since  $P(t) = \frac{1}{2}$  for  $t \geq 1$ .
- If  $p \leq \frac{1}{3}$ ,  $P(t - 1) - R(t) = [p^{t-1}P(0) - p^{t-1}P(1) + \text{other positive factors}] = p^{t-1}q + \epsilon$ .

$$t = 3, P(3) - R(4) > pq^{4-1}P(4-1) \quad (11)$$

$$p^3P(0) - p^3P(1) \quad (12)$$

$$\binom{3}{2}p^2qP(1) - \binom{3}{2}p^2qP(2) \quad (13)$$

$$\binom{3}{1}p^2qP(2) - \binom{3}{2}p^2qP(3) \quad (14)$$

$$(15)$$